

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (CE)

May 2012

CE305 Computer Networks & Internetworking Layers

Date: 07.05.2012, Monday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q – 1 Answer the following questions.

- (a) What are the two reasons for using layered protocols? What is the principle difference between connectionless communication and connection oriented communication? [04]
- (b) What is the difference between error correction and error detection? Define ARQ (Automatic Repeat Request). [03]

Q – 2 Answer the following questions.

- (a) What is the remainder obtained by dividing $x^7 + x^5 + 1$ by the generator polynomial $x^3 + 1$? And explain the process on the receiving end to calculate CRC. [05]
- (b) Explain HDLC frame format in detail. [05]
- (c) What is the difference between Pure Aloha and Slotted Aloha multiple access protocol? Explain both protocols in brief. [04]

OR

Q – 2 Answer the following questions.

- (a) A bit stream 10011101 is transmitted using the standard CRC method. The generator polynomial is $x^3 + 1$. Show the actual bit string transmitted. Suppose the third bit from the left is inverted during transmission. Show that this error is detected at the receiver's end. [05]
- (b) Explain Go back-N protocol of frame control in details. And what is the windows size on sender's side and receiver side in Go back-N protocol. [05]
- (c) Explain the hidden station problem and exposed station problem of wireless LAN for multiple access. [04]

Q - 3 Answer the following questions.

- (a) Explain the optimality principle. What are the difference between adaptive routing algorithm and non-adaptive routing algorithm? List the entire adaptive and non-

adaptive algorithm.

- (b) Compare and contrast the OSI and TCP/IP reference models. [05]
- (c) Explain carrier sense multiple access protocols in detail. [05]

OR

Q - 3 Answer the following questions.

- (a) Explain shortest path routing and flooding routing algorithm in brief. [04]
- (b) List all the services provided by all the layers of OSI reference model. [05]
- (c) Explain wavelength division multiple access protocols in detail. [05]

SECTION - II

Q - 4 Answer the following questions.

- (a) Compare classfull IP address Vs classless IP address of IPv4. [03]
- (b) Explain the addressing performed at transport layer in detail. [04]

Q - 5 Answer the following questions.

- (a) Explain distance vector routing algorithm in brief. Explain count to infinity problem occurred in distance vector routing in brief. [04]
- (b) Explain various techniques to deal with congestion control in datagram subnet. [05]
- (c) Draw and explain each field of TCP header in detail. [05]

OR

Q - 5 Answer the following questions.

- (a) Explain link state routing algorithm in brief and compare link state algorithm with distance vector routing algorithm. [04]
- (b) Explain congestion prevention policies at transport, network and data link layer. [05]
- (c) Explain multiplexing and crash recovery at the transport layer. Explain various transport service primitives in brief. [05]

Q - 6 Answer the following questions.

- (a) Explain DNS (Domain Name System) in detail. [07]
- (b) Explain connection release process of TCP in detail with all four cases. [07]

OR

Q - 6 Answer the following questions.

- (a) Explain Electronic mail in detail. [07]
- (b) Explain flow control and buffering techniques used at transport layer. [07]
